FCAL: Foreground-Constrained Attention Learning for Background-Robust Classification

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***Abstract*—We introduce Foreground-Constrained Attention Learning (FCAL), a mask-supervised training objective for classification under background change. Its methodological contribution couples fractional positive-evidence containment, scale-normalized signed foreground support, and resolution-aware boundary tolerance on an established linear-head evidence map. FCAL requires one backbone forward pass and adds no inference operations or parameters. We provide a fixed starting recipe and evaluate six architectures on CUB-200-2011 and Oxford-IIIT Pet. Across three matched CUB ResNet-50 seeds, FCAL improves clean accuracy by 1.05 ± 0.19 percentage points and mean background-intervention accuracy by 3.65 ± 1.37 points over cross-entropy; the reported spreads are sample standard deviations of paired seed changes. In the single-seed six-architecture comparison, fixed FCAL has higher mean accuracy than fixed CAMAL on all four reported classification endpoints, whereas CAMAL leads common-map localization and tuned CNN worst-background accuracy. The architecture matrix shows stronger background gains over cross-entropy for CNNs than for average-pooled ViTs. A separate 100-image ISIC negative check illustrates the transfer boundary. FCAL offers a concrete method for preserving classification accuracy while improving background robustness, with the strongest replicated evidence on CUB ResNet-50.**

***Index Terms*—explanation-guided learning, background robustness, class activation maps, foreground supervision, convolutional neural networks, vision transformers**

I. Introduction

Image classifiers can exploit background cues that correlate with labels during training but change at deployment: a spatial form of shortcut learning [1]. A bird's habitat or a pet's domestic setting can supply class signal while creating background sensitivity that differs across architectures [2], [3]. Clean accuracy alone therefore does not characterize background robustness.

Foreground masks offer a way to influence training: penalize model evidence outside a region considered relevant to the task. Explanation-guided learning already includes input-gradient regularization, trainable attention, and relevance-map optimization [4], [5], [6]. FCAL addresses a specific objective-design problem: coupling evidence containment with foreground support at a consistent spatial tolerance, without requiring additional backbone passes. Its purpose is to preserve classification accuracy while improving predictions under background change.

Three design issues motivate FCAL. *Scale:* raw activation magnitudes depend on features and classifier weights, so an outside-activation penalty does not directly measure a background fraction. *Sign:* a fully non-positive map has zero positive outside mass and supplies no containment gradient. *Resolution:* fixed-cell dilation produces different input-space tolerances on coarse CNN maps and finer ViT grids. Training cost also matters: input-gradient differentiation and attention-mining passes increase runtime and memory [4], [5], [7].

We propose FCAL to address these issues jointly. Its containment term penalizes normalized positive evidence outside a tolerance-expanded foreground. A bounded signed-support term encourages positive scores across the strict foreground, including when the positive-map penalty supplies no gradient. The tolerance is defined in input pixels using the evidence-grid resolution. Both terms operate on features from the ordinary classification forward pass. The CNN/ViT pairing is deliberate: final convolutional features and normalized ViT patch tokens admit the same linear score decomposition when their classification head uses average pooling. This supplies a common algebraic interface for the proposed objective across both architecture families.

We evaluate a fixed FCAL recipe with clean-validation checkpoint selection, then test coefficient sensitivity separately. The experiments compare its predictive behavior, spatial evidence, transfer, and computational cost with matched controls and explanation-supervision baselines.

The contributions are threefold:

- **A coupled foreground-supervision objective.** FCAL combines fractional containment, signed support, and input-space boundary tolerance on a shared CNN/average-ViT evidence interface, with unchanged inference.
- **An explicit training recipe with replicated gains.** The fixed configuration improves clean, background-intervention, and foreground-only accuracy across three matched CUB ResNet-50 seeds; a component ablation examines the roles of tolerance and support.
- **A measured operating advantage and transfer boundary.** Fixed and validation-tuned comparisons identify when FCAL is preferable to CAMAL, GAIN-ext, RRR, and the supported Faster-ISNet implementation, while the CNN/ViT matrix and external checks establish the scope of transfer.

Section II positions FCAL within related work. Section III separates the established evidence decomposition from the proposed objective. Section IV presents the experimental protocol and results. Section V discusses mechanisms and limitations, Section VI describes reproducibility, and Section VII concludes.

II. Related Work

FCAL connects spatial explanations, explanation-supervised training, and background-robust classification.

A. Spatial evidence and transformer explanations

Class activation mapping (CAM) projects final spatial features through a linear classifier after global average pooling [8]. Grad-CAM extends localization to more general networks through class gradients [9]. Saliency maps can fail model-sensitivity checks [10]; we distinguish their spatial alignment from prediction robustness and causal pixel importance.

Chefer et al. [11] optimize ViT relevance maps to suppress background relevance. ICEv2 (*IJCV*, 2025) uses patch-wise classification and tailored training for explainability visualization [12]. FCAL instead uses a common linear-head interface for GAP CNNs and ViTs with averaged normalized patch tokens.

B. Explanation-guided training

Gao et al.'s 2024 survey [13] reviews explanation supervision, regularization, and intervention. RRR penalizes input gradients in annotated irrelevant regions [4]. GAIN makes gradient-derived attention trainable and combines classification with attention mining; its externally supervised form uses spatial annotations [5]. HINT aligns human and model importance in vision-language tasks [14].

ISNet minimizes background Layer-wise Relevance Propagation (LRP) relevance to reduce background bias [6]. Faster ISNet removes the original method's class-count dependence in training and introduces LRP-Flex [7]. Both retain the inference graph. Our supported implementation comparisons use a shared CUB protocol.

CAMAL rewards foreground CAM activation and suppresses background activation using segmentation masks [15]. At the time of writing, it is available as a 2026 arXiv preprint rather than an archival peer-reviewed publication. We include it because its regional-mean objective and unchanged inference cost make it the closest contemporary comparator to FCAL. The comparison uses shared splits and fixed or validation-selected recipes, not CAMAL's original cross-validation, and is interpreted as descriptive evidence.

FCAL's methodological contribution couples evidence-scale normalization, signed foreground support, and grid-aware input-space tolerance in one objective. CAM supplies its established evidence interface.

C. Evaluation scope

Our region-preserving tests build on background-sensitivity studies [2], [3] and corruption benchmarking [16]. This custom suite measures controlled interventions, not a natural deployment distribution. A common CAM estimator separates training-method comparisons from explainer choice; localization and insertion/removal are reported separately from accuracy.

III. FCAL: Method and Training Recipe

FCAL uses class labels and foreground masks during training. For one image, let $x \in \mathbb{R}^{3\times H\times W}$ be the input, $y \in \{1,\dots,K\}$ its class, and $M \in \{0,1\}^{H\times W}$ its foreground mask. Here H, W are input dimensions and K is the number of classes. The network emits logits $z \in \mathbb{R}^K$. We first specify the established map used by the objective, then define the proposed loss components. All spatial sums below run over the $h\times w$ evidence grid unless otherwise stated.

A. Established foundation: linear-head class evidence

Let $A_{ij} \in \mathbb{R}^D$ be the final feature vector at spatial cell (i,j), and let $w_k \in \mathbb{R}^D$ and $b_k \in \mathbb{R}$ be the weight and bias of class k. Equation (1) defines the class evidence and the corresponding GAP logit identity:

$$S_{ij}(x,k) = w_k^\top A_{ij}, \qquad z_k(x) = \frac{1}{hw}\sum_{ij} S_{ij}(x,k) + b_k. \quad (1)$$

For a CNN, A_{ij} is the final convolutional feature vector. For the evaluated ViTs, it is the final normalized patch token; prefix tokens are excluded from pooling and no nonlinear normalization follows the spatial average. Thus the same identity holds for both families. EfficientNet's training map reuses the classifier's sampled channel-dropout scale in A_{ij}, rather than sampling a separate mask. At evaluation, dropout is disabled. These choices preserve the score identity, subject to floating-point precision, without interpreting raw self-attention as CAM. The identity is inherited from the head structure [8]; FCAL's proposed contribution is the objective applied to it.

B. Proposed component: fractional containment with spatial tolerance

Training uses the ground-truth-class map $S = S(x,y)$. Equation (2) defines its non-negative spatial mass:

$$P_{ij} = \frac{\mathrm{ReLU}(S_{ij})}{\sum_{uv}\mathrm{ReLU}(S_{uv}) + \epsilon}, \qquad \epsilon = 10^{-8}. \quad (2)$$

The indices u, v range over the same evidence grid as i, j. For positive total evidence well above ϵ, P approximately sums to one and makes an outside penalty interpretable as a fraction. The stabilized expression is approximately invariant to positive rescaling away from the numerical floor; it is not an exactly normalized probability distribution when the total mass is tiny. If all scores are non-positive, P is identically zero.

Boundary tolerance accounts for the grid resolution. Equation (3) converts a fraction τ of an evidence cell into a capped input-pixel radius δ:

$$\delta = \min(\delta_{\max}, \lceil \tau \max(H/h, W/w) \rceil). \quad (3)$$

With $\tau = 0.5$ and $\delta_{\max} = 24$ pixels, a 224×224 input gives a radius of 16 pixels on a 7×7 grid and 8 pixels on a 14×14 grid. Let $\mathcal{A}_{h,w}$ denote adaptive average pooling to the evidence grid and $\mathcal{D}_\delta$ denote input-grid dilation using a square window of side $2\delta + 1$. Equation (4) defines strict and tolerant soft foreground weights:

$$G = \mathcal{A}_{h,w}(M), \qquad \widetilde{G} = \mathcal{A}_{h,w}(\mathcal{D}_\delta(M)). \quad (4)$$

Dilation precedes projection; it is not a fixed number of pooled-map cells. The proposed per-image containment loss is then

$$\ell_{\text{out}} = \sum_{ij} P_{ij}(1 - \widetilde{G}_{ij}). \tag{5}$$

Equation (5) constrains distant positive evidence while allowing a boundary shell. It does not require uniform activation over the object, and it does not penalize negative background scores.

C. Proposed component: signed foreground support

Containment alone gives no spatial gradient through a fully non-positive ReLU map. We therefore retain signed scores in a complementary support term. Equation (6) defines a per-image RMS scale r and normalized score $\widehat{S}$, where sg denotes stop-gradient:

$$r = \text{sg}\left(\sqrt{\frac{1}{hw}\sum_{ij} S_{ij}^2 + \epsilon}\right), \qquad \widehat{S}_{ij} = S_{ij}/r. \tag{6}$$

The strict foreground weights G then define bounded mean support and its penalty in Equation (7):

$$\begin{aligned} q_{\text{fg}} &= \frac{\sum_{ij} G_{ij} \tanh(\widehat{S}_{ij}/T)}{\max(\sum_{ij} G_{ij}, \epsilon)}, \\ \ell_{\text{sup}} &= \text{softplus}(m - q_{\text{fg}}). \end{aligned} \tag{7}$$

Here $T = 1$ is a temperature, $m = 0$ is a support offset, and $\text{softplus}(a) = \log(1 + e^a)$. A cell's bounded contribution is limited by its foreground weight; increasing one activation without bound cannot produce an unbounded increase in q_{fg}. This encourages distributed support without enforcing equal evidence at all object locations. The term can supply a score gradient when positive containment is inactive, but saturation and network optimization prevent a guarantee that every trained map becomes positive. Stop-gradient intentionally supplies a surrogate gradient through the normalized scores. Also, softplus remains active when $q_{\text{fg}} > m$; the offset is not a hard stopping threshold.

D. Combined objective and default recipe

Let $\mathcal{L}_{\text{CE}}$ be mean multiclass cross-entropy over the mini-batch, and let $\mathcal{L}_{\text{out}}$ and $\mathcal{L}_{\text{sup}}$ average the corresponding per-image losses over valid masks. Cropped masks that are entirely foreground or background are excluded from spatial losses but remain classification samples; an empty valid set contributes zero spatial loss. Equation (8) defines the detached adaptive containment weight:

$$\lambda = \text{clip}\left(\alpha \frac{\text{sg}(\mathcal{L}_{\text{CE}})}{\text{sg}(\mathcal{L}_{\text{out}}) + \epsilon}, \lambda_{\min}, \lambda_{\max}\right). \tag{8}$$

The scale factor is $\alpha = 0.15$, the lower bound is $\lambda_{\min} = 0.05$, and the upper bound is 0.30 for CNNs and 1.00 for ViTs. Clipping prevents unrestricted compensation when containment is small. With epoch number t starting at one, five warm-up epochs and a five-epoch linear ramp give $\rho_t = \min(1, \max(0, (t-5)/5))$. Equation (9) specifies the complete objective:

$$\mathcal{L}_{\text{FCAL}} = \mathcal{L}_{\text{CE}} + \rho_t \left(\lambda \mathcal{L}_{\text{out}} + \beta \mathcal{L}_{\text{sup}}\right), \qquad \beta = 0.05. \tag{9}$$

Starting configuration. We recommend the fixed $\beta = 0.05$ recipe in Equation (9) as the initial FCAL configuration. It is the original declared setting used for the primary matrix and three-seed replication. The tuned arm varies β only and is reported as a coefficient-selection sensitivity study, not a replacement default. All other FCAL settings remain fixed. This prescription retains the original recipe rather than selecting a different coefficient for each test endpoint.

FCAL projects one selected classifier row onto cached spatial features and computes mask-weighted reductions. The evidence projection costs $O(Dhw)$ per image, with no extra backbone pass and no differentiation through input gradients. It uses the original training input after the shared data augmentation, without FCAL-specific image interventions. Masks, map extraction, and auxiliary losses are discarded at inference; the deployed architecture and parameter count equal those of the matched CE model.

IV. Experiments and Results

We test FCAL's classification performance under controlled background changes, its transfer across architectures and datasets, and its cost against explanation-supervision baselines. The protocol separates the replicated ResNet-50 comparison from the single-seed architecture matrix.

A. Datasets, architectures, and training

CUB-200-2011 contains 200 bird classes with segmentation masks [17]; our split has 4,794 training, 1,200 validation, and 5,794 official test images. Oxford-IIIT Pet contains 37 breeds and trimaps [18]; 2,944/736 images form the training/validation partition of trainval, with 3,669 official test images held out. CUB and Pet are the primary classification benchmarks. A separate ISIC negative check [19] uses eight classes with group-disjoint 1,414/356/100 training/validation/test images; its rarest test class has three images. This small subset probes a transfer failure rather than establishing clinical effectiveness.

The six ImageNet-pretrained backbones are ResNet-50, DenseNet-121, EfficientNet-B0, and average-pooled ViT-Tiny/16, ViT-Small/16, and ViT-Base/16 [20]. Inputs are 224×224. Primary runs train for 50 epochs with AdamW, batch size 32, weight decay 10^{-4}, cosine learning-rate decay to 1% of the initial rate, and bfloat16 mixed precision. Initial learning rates are 2×10^{-4} for CNNs and 5×10^{-5} for ViTs. Image and mask spatial transformations are paired; evaluation uses deterministic resizing and center cropping. Checkpoints are selected by validation accuracy.

For each dataset, architecture, and training seed, the CE run saves the model, optimizer, scheduler, and random-state information after epoch 5. FCAL resumes that branch and completes the same schedule. This matches initialization and the saved data-order state; different objectives can subsequently consume random numbers differently. The CE/FCAL matrix

contains 24 primary CUB/Pet models and 12 ISIC-check models: two objectives and six architectures per dataset, all at seed 42. CUB ResNet-50 additionally uses training seeds 7 and 123 on the same fixed split, giving three matched pairs.

B. Comparator and selection protocols

The fixed CUB arm includes CAMAL, GAIN-ext, and RRR on all six architectures at seed 42. RRR uses the summed-log-probability input-gradient penalty; GAIN-ext includes supervised target attention and its attention-mining forward; CAMAL uses foreground/background CAM alignment. The implemented selective Faster-ISNet/LRP-Flex path is audited on ResNet-50 only. All use the corresponding epoch-5 CE branch and validation checkpoint rule. Fixed refers to the declared project recipes. FCAL uses the configuration defined in Section III.

A validation-only successive-halving study searches five predeclared values of one coefficient per supported method–architecture cell. At 10, 25, and 50 total epochs it retains $5 \to 2 \to 1$ candidates, each restarted from the same epoch-5 state. The 200 jobs produce 25 winners frozen before test evaluation. The search matches dimensionality and epoch budgets, while method runtimes differ. Appendix A gives the grids, selected values, and cost.

C. Regional, explanation, and cost measurements

The common evaluation has 30 deterministic views: clean and sham controls; 13 background and 13 foreground interventions; and foreground-only and background-only ablations. Each intervention family contains three blur, three color-jitter, and three Gaussian-noise severities, plus grayscale, solid mean, random solid color, and random RGB. For a background intervention, foreground pixels are copied exactly from the original, and conversely for foreground interventions. Stochastic transformations are keyed by seed, image identifier, and variant. Region-only ablations replace the removed region with the neutral mean color.

BG mean averages test accuracy over the 13 background views; BG worst is their minimum dataset-level accuracy for each model. FG mean/worst are analogous. Worst values are computed before architecture averaging. FG only measures object sufficiency under mean-color replacement; BG only measures residual class signal.

Strict leakage applies Equation (2) to the ground-truth-class map with undilated grid mask G:

$$\ell_{\text{strict}} = \sum_{ij} P_{ij}(1 - G_{ij}). \tag{10}$$

Lower values indicate less positive map mass outside the strict mask. This spatial metric is unconditional, assigns zero to an empty positive map, and does not measure causal dependence.

Common-map faithfulness uses the same 500 test images, selected by a label-independent image-ID hash with selection seed 2026, for every CUB checkpoint. Non-negative ground-truth CAMs are max-normalized and bilinearly upsampled. Mask IoU thresholds them strictly above 0.7. Insertion/removal ranks pixels by this map and evaluates ground-truth softmax confidence at fractions $0, 0.1, \ldots, 1$; trapezoidal AUC uses a zero normalized-input baseline, equivalent to the ImageNet mean color. Higher insertion and lower removal AUC are favorable. These scores compare a common map rather than native RRR gradients or ISNet relevance maps.

Training cost is measured on one NVIDIA GeForce RTX 5070 Ti. Runtime sums training and validation for epochs 6–50, excluding the shared warm-up; memory is peak allocated CUDA memory. Search cost charges every tuning job.

D. Results: predictive gains of the fixed recipe

The fixed FCAL recipe improves all four predictive endpoints over CE in each of three matched CUB ResNet-50 seeds (Table I). Clean accuracy increases by 1.05 ± 0.19 points, BG mean by 3.65 ± 1.37, and FG only by 3.87 ± 0.57. BG-worst accuracy gains 12.77 ± 6.63 points, showing greater seed sensitivity than the clean endpoint. Figure 1 displays the individual paired changes. Mean strict leakage decreases from 0.571 to 0.281. These results establish the main practical finding: the declared FCAL recipe improves background robustness without sacrificing clean accuracy in the replicated comparison.

The full dataset averages, including foreground-corruption and background-only diagnostics, are retained in Appendix B. The smaller CUB FG-corruption mean shows that the improvement is background-oriented rather than uniform resistance to damage anywhere in the image.

E. Results: transfer across architectures

FCAL's background benefit is strongest for the evaluated CNNs. Table II separates the three CNNs from the three ViTs. On CUB, CNN mean BG accuracy increases by 2.48 points and BG-worst accuracy by 9.00 points. The corresponding ViT changes are -0.06 and $+0.29$ points. Yet leakage decreases by 0.207 for CNNs and 0.216 for ViTs. The shared objective therefore improves containment in both family means while delivering a much larger predictive gain for CNNs.

Architecture-level results sharpen this interpretation (Table III). ResNet-50 supplies the largest CUB gain: 5.19 points in BG mean and 20.38 in BG worst. DenseNet and EfficientNet also improve both endpoints, although EfficientNet loses 1.19 points in FG-only accuracy. ViT-Tiny and ViT-Small slightly regress on both background endpoints, whereas ViT-Base improves. The ViT leakage mean is strongly influenced by ViT-Base's 0.618 reduction; Tiny and Small decrease by only 0.017 and 0.015.

F. Results: direct comparison with CAMAL

Table IV separates selection regimes and architecture families. Across all six architectures, fixed FCAL exceeds fixed CAMAL by 1.02 points clean, 1.27 BG mean, 0.93 BG worst, and 1.44 FG only. It leads these endpoints on 6/6, 5/6, 4/6, and 6/6 models, respectively, at similar training cost.

Within CNNs, FCAL gains 1.09 points clean, 1.27 BG mean, and 1.02 FG only, with BG worst lower by 0.09 points.

TABLE I

CUB RESNET-50 TEST ACCURACY ACROSS TRAINING SEEDS 7, 42, AND 123. VALUES ARE MEAN ± SAMPLE SD; THE LAST ROW SUMMARIZES PAIRED SEED DIFFERENCES IN PERCENTAGE POINTS.

Method	Clean	BG mean	BG worst	FG only
CE	77.44±0.13	68.75±1.70	38.90±9.81	68.63±0.34
FCAL (proposed)	78.49±0.16	72.40±0.38	51.67±4.00	72.51±0.87
Δ FCAL–CE	+1.05±0.19	+3.65±1.37	+12.77±6.63	+3.87±0.57

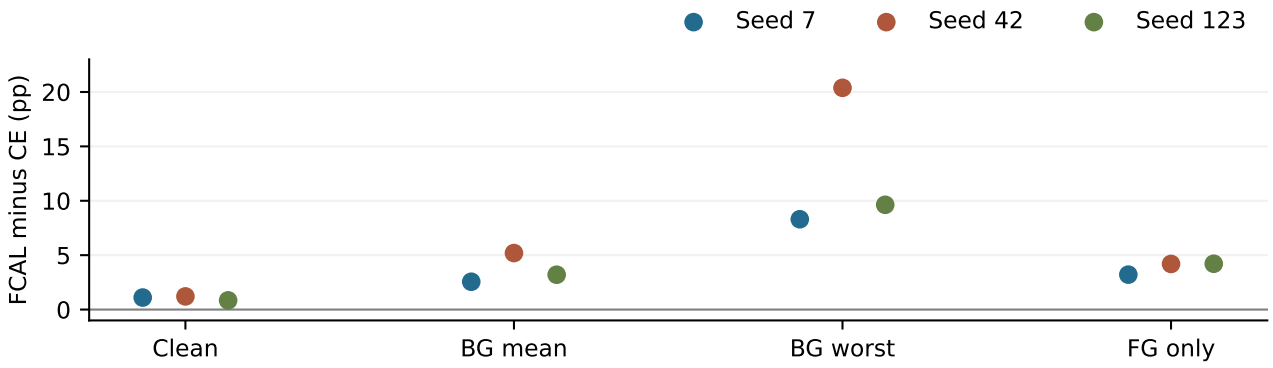


Fig. 1. Individual paired CUB ResNet-50 seed differences on the test set. All four accuracy endpoints favor FCAL in these three pairs. BG-worst gains vary substantially by seed; points are training repetitions on one fixed data split, not independent datasets.

TABLE II

FIXED-RECIPE TEST CHANGES, FCAL MINUS CE, WITHIN EACH ARCHITECTURE FAMILY (SEED 42). CUB AND PET ARE PRIMARY BENCHMARKS; ISIC IS A SMALL NEGATIVE CHECK. ACCURACY CHANGES ARE PERCENTAGE POINTS; LEAKAGE IS A FRACTION. EACH ROW AVERAGES THREE ARCHITECTURES. BG WORST IS COMPUTED PER ARCHITECTURE BEFORE AVERAGING.

Dataset	Family	Δ Clean	Δ BG mean	Δ BG worst	Δ FG only	Δ Leak ↓
CUB	CNN	+0.81	+2.48	+9.00	+1.64	-0.207
CUB	ViT	+0.34	-0.06	+0.29	+0.21	-0.216
Pet	CNN	+0.55	+0.77	+1.95	+4.13	-0.070
Pet	ViT	+0.45	+0.57	+0.65	+0.58	-0.132
ISIC	CNN	-1.67	+1.18	-1.67	+7.00	-0.138
ISIC	ViT	-1.67	-0.85	-0.33	+0.33	-0.229

TABLE III

CUB FIXED-RECIPE TEST CHANGES BY ARCHITECTURE, FCAL MINUS CE (SEED 42). ACCURACY CHANGES ARE PERCENTAGE POINTS; LEAKAGE CHANGES ARE FRACTIONS. THIS DECOMPOSITION EXPOSES HETEROGENEITY HIDDEN BY FAMILY MEANS.

Architecture	Δ Clean	Δ BG mean	Δ BG worst	Δ FG only	Δ Leak ↓
ResNet-50	+1.21	+5.19	+20.38	+4.19	-0.287
DenseNet-121	+0.14	+1.35	+4.38	+1.92	-0.217
EfficientNet-B0	+1.07	+0.89	+2.24	-1.19	-0.116
ViT-Tiny/16 avg	+0.52	-0.49	-1.05	-0.19	-0.017
ViT-Small/16 avg	-0.43	-0.44	-0.14	-0.38	-0.015
ViT-Base/16 avg	+0.93	+0.75	+2.07	+1.21	-0.618

DenseNet-121 improves all four endpoints over CAMAL by 1.54, 2.28, 3.71, and 2.26 points, respectively. EfficientNet-B0 also improves all four, including a 6.61-point BG-worst gain.

The comparison changes with the backbone and endpoint. On fixed ResNet-50, FCAL gains 0.57 points clean over CAMAL but loses 0.70 in BG mean and 10.58 in BG worst. CAMAL also provides higher common-map IoU and insertion AUC, as shown below. Section V converts these endpoint-level results into a practical method choice.

Across all six architectures, fixed FCAL has higher clean accuracy than CAMAL and RRR on every model and than GAIN-ext on five. It has the highest CNN mean clean, BG-mean, and FG-only accuracy among the listed fixed methods; RRR has the highest CNN BG-worst mean. For ViTs, FCAL improves clean and FG-only means over CE, while CE retains a slightly higher BG mean.

TABLE IV

CUB FIXED AND VALIDATION-TUNED COMPARISONS (SEED 42). THE FIRST FIVE NUMERIC COLUMNS AVERAGE THE THREE CNNS; THE FINAL FIVE AVERAGE THE THREE VITS. ACCURACY IS PERCENT; LEAKAGE IS A FRACTION. THE SAME FIXED CE REFERENCE APPLIES TO BOTH SELECTION REGIMES.

Arm	Method	CNN Clean	CNN BG mean	CNN BG worst	CNN FG only	CNN Leak ↓	ViT Clean	ViT BG mean	ViT BG worst	ViT FG only	ViT Leak ↓
Fixed	CE	78.01	68.97	36.66	69.66	0.566	80.95	76.10	64.13	75.12	0.803
Fixed	FCAL	78.82	71.45	45.67	71.30	0.360	81.29	76.04	64.42	75.33	0.587
Fixed	CAMAL	77.72	70.19	45.75	70.28	0.345	80.35	74.77	62.48	73.46	0.439
Fixed	GAIN-ext	70.87	61.46	31.81	62.74	0.237	80.77	75.88	64.66	74.84	0.523
Fixed	RRR	70.94	66.70	55.43	65.97	0.531	78.45	75.16	66.32	74.62	0.801
Tuned	FCAL	78.58	70.43	41.31	70.85	0.367	81.26	75.95	63.69	75.11	0.566
Tuned	CAMAL	78.18	71.28	51.95	68.83	0.391	80.49	75.03	61.98	73.74	0.483
Tuned	GAIN-ext	71.17	60.97	26.55	62.10	0.203	80.94	75.61	64.22	74.18	0.404
Tuned	RRR	73.54	68.68	54.61	68.77	0.542	79.78	76.29	66.61	75.99	0.802

After validation tuning, FCAL retains the highest six-architecture mean clean and FG-only accuracy among the four fully supported regularizers. Its BG-mean result is a descriptive near-tie with CAMAL: 73.19% versus 73.16%. CAMAL and RRR achieve higher BG-worst means, and GAIN-ext gives the lowest leakage. In the tuned CNN arm, FCAL retains higher clean and FG-only accuracy than CAMAL on all three backbones, while CAMAL has higher BG mean and BG worst on all three. The same priority-based choice remains relevant, with a larger robustness advantage for tuned CAMAL.

The coefficient search does not supply a consistently better FCAL configuration. Relative to the fixed recipe, FCAL's validation-selected recipe has lower clean, BG-mean, and BG-worst test accuracy on five of six architectures. FG-only accuracy decreases on four, ties on one, and increases on one. RRR improves clean, BG mean, and FG-only accuracy on all six. Clean-validation selection therefore affects the methods differently and does not directly optimize background robustness. We retain the originally declared FCAL recipe as the reproducible default; the tuned arm measures selection sensitivity. On a new task, assess a different coefficient against that default using a held-out validation endpoint chosen for the application. The reported test outcomes support this starting prescription, not a proof that tuning cannot help on other data.

The Faster-ISNet comparison is restricted to its supported ResNet-50 pairing (Table V). The fixed pairing favors FCAL on all listed metrics, including lower leakage. In the tuned pairing, FCAL remains higher on clean, BG mean, and FG-only accuracy, while Faster-ISNet has higher BG-worst accuracy. Its single-architecture result is never included in a six-architecture mean.

G. Results: spatial alignment and common-map faithfulness

Figure 2 shows selected clean-image examples across the three datasets. Figure 3 shows selected CUB ResNet-50 cases where CE is wrong and FCAL is correct under clean or background-modified inputs. The captions identify their selection and map conventions; quantitative comparisons use the full evaluation sets. For localization and insertion AUC, CAMAL records the higher values. FCAL improves the fixed six-architecture mean IoU, insertion AUC, and removal AUC over CE, but CAMAL has higher IoU and insertion AUC in both arms. GAIN-ext has lower removal AUC than FCAL in both arms. At architecture level, CAMAL's IoU exceeds FCAL's on all six models, and GAIN-ext's removal AUC is lower on all six, in both selection regimes (Table VI).

These outcomes distinguish the objectives in practice: FCAL's predictive benefit need not coincide with the most spatially complete map. The AUC comparison uses the common-CAM convention and mean-color replacement specified above.

H. Results: training and selection cost

Table VII shows that FCAL trains faster than the evaluated GAIN-ext, RRR, and Faster-ISNet implementations. CAMAL is the closest efficient competitor, with nearly identical memory use and a 1.09 median time ratio. RRR's wide cross-architecture range reflects implementation-specific costs.

For Faster-ISNet on ResNet-50, the post-warm-up time ratio is 9.44 and the peak-memory ratio is 6.16 relative to FCAL. Including the shared warm-up instead gives a 7.55 total-history time ratio; these denominators describe different measurements. The primary table consistently uses the former. FCAL's training adds work, while its inference cost is unchanged by construction, as it is for CAMAL and the other training-only regularizers.

The 200 tuning jobs used 21.604 allocated GPU-hours: 2.287 for FCAL, 2.326 for CAMAL, and 11.566 for RRR. Appendix A gives the complete accounting.

I. Results: component ablation

The controlled component study tests the two complementary design choices within FCAL (Table VIII). Each row uses CUB ResNet-50, seed 42, the same epoch-5 branch, and the epoch-20 endpoint while retaining the 50-epoch cosine schedule. The fixed-cell variant keeps signed support but dilates the projected foreground by three evidence cells. The no-support variant retains resolution-aware tolerance and sets $\beta = 0$.

Replacing fixed-cell tolerance with the proposed tolerance increases BG-worst accuracy by 8.25 points at nearly unchanged clean accuracy. Adding support to the tolerance-only objective increases BG-worst accuracy by 8.63 points, with a 0.64-point clean-accuracy reduction. Thus the full objective is favored for the damaging-background endpoint, while the no-support variant retains a small clean advantage. The study supports

TABLE V
SUPPORTED RESNET-50-ONLY CUB COMPARISON, SEED 42. THESE VALUES ARE NOT SIX-ARCHITECTURE MEANS.

Arm	Method	Clean	BG mean	BG worst	FG only	Leak ↓
Fixed	FCAL (proposed)	78.49	72.11	48.86	72.76	0.282
Fixed	Faster-ISNet	72.06	66.16	47.05	67.67	0.547
Tuned	FCAL (proposed)	78.27	71.52	45.46	72.58	0.245
Tuned	Faster-ISNet	74.77	68.77	50.55	69.26	0.537

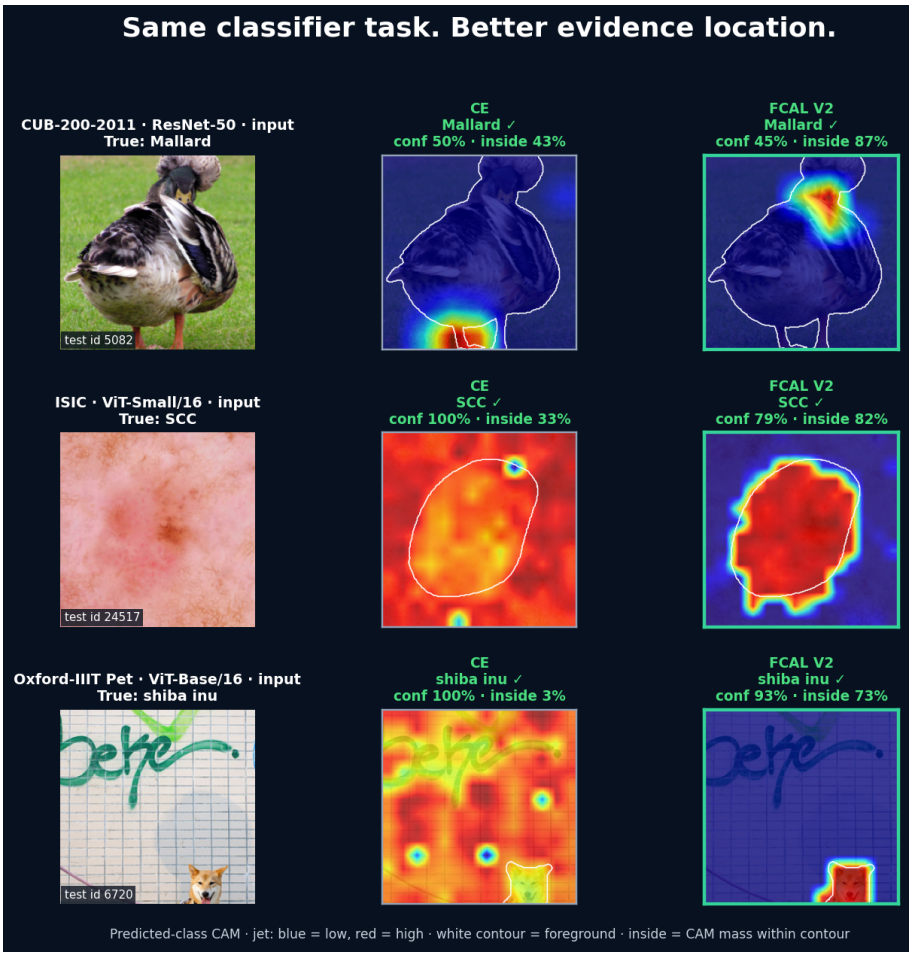


Fig. 2. Selected clean-image evidence maps from CUB, ISIC, and Oxford Pet. The white contour marks the annotated foreground. Both classifiers are correct in these examples; the displayed positive evidence is more foreground-contained for FCAL. Maps target each model's predicted class, whereas numerical leakage uses the ground-truth class. The original panel label "FCAL V2" denotes FCAL in this article. Selected examples do not summarize dataset-level accuracy or causal dependence.

these design choices for one architecture and seed; it is not a factorial interaction estimate or a demonstration that all positive maps remain live.

J. Results: Pet transfer and the ISIC negative check

Oxford Pet provides positive descriptive transfer: five of six architectures improve clean accuracy and all six reduce leakage. Exact two-sided McNemar tests compare each fixed FCAL model with its paired CE model on clean test predictions. After Holm correction across the six architecture comparisons, only ViT-Small remains significant at 0.05, favoring FCAL ($p_{\text{raw}} = 0.000218$, $p_{\text{Holm}} = 0.00131$). The borderline ViT-Tiny decline ($p_{\text{raw}} = 0.0499$) does not survive correction ($p_{\text{Holm}} = 0.249$). These tests are conditional on the trained pairs and do not quantify training-seed variability. The family decomposition also favors CNNs on FG-only transfer: the Pet mean gain is 4.13 points for CNNs versus 0.58 for ViTs. Both families improve BG mean, by 0.77 and 0.57 points respectively (Table II).

The small ISIC check supplies a negative transfer result: six-architecture mean clean accuracy decreases from 66.50% to 64.83%, despite lower leakage and higher FG-only accuracy. BG-worst accuracy also decreases. The 100-image test set,

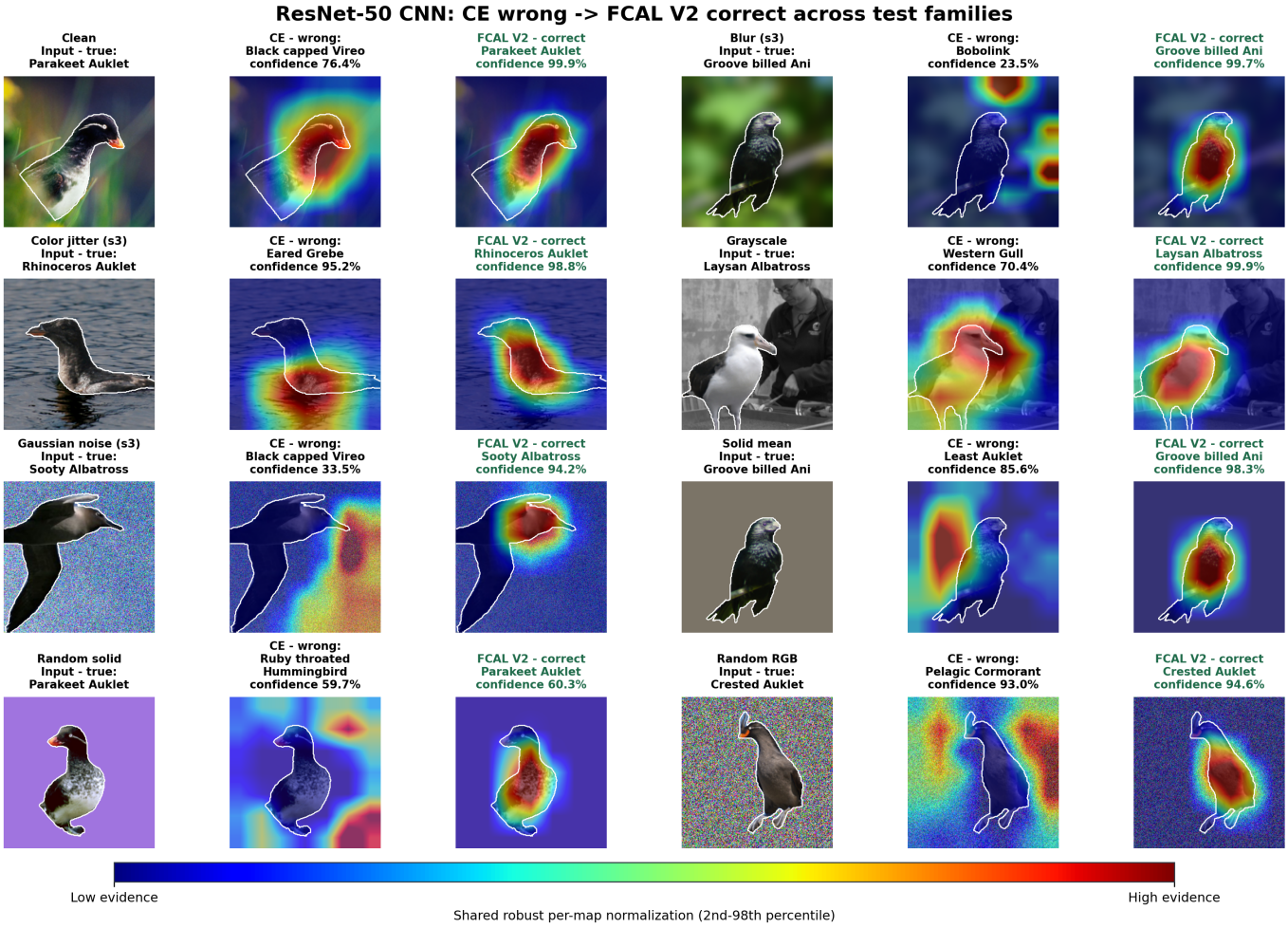


Fig. 3. Selected CUB ResNet-50 prediction rescues: one clean example and seven background-intervention conditions, with foreground pixels preserved in each intervention. Each triplet shows the input, CE map, and FCAL map; examples can differ across conditions. Maps target the displayed predicted class, which differs between the models in these selected cases. Colors use per-map 2nd–98th percentile normalization and cannot compare absolute evidence magnitudes across panels. "FCAL V2" is the retained FCAL method; "s3" is the highest tested severity. These illustrations are selected successes, not an unbiased robustness sample.

TABLE VI
COMMON-CAM EVALUATION ON THE SAME 500 CUB TEST IMAGES FOR EVERY MODEL. ALL ENTRIES ARE SIX-ARCHITECTURE MEANS. HIGHER IOU AND INSERTION AUC (INS.) AND LOWER REMOVAL AUC (REM.) ARE FAVORABLE; AUCS USE GROUND-TRUTH SOFTMAX CONFIDENCE. CE HAS NO TUNED ARM.

Method	Fixed IoU ↑	Ins. ↑	Rem. ↓	Tuned IoU ↑	Ins. ↑	Rem. ↓
CE	0.239	0.388	0.363	–	–	–
FCAL (proposed)	0.358	0.499	0.247	0.368	0.557	0.191
CAMAL	0.667	0.641	0.097	0.623	0.644	0.096
GAIN-ext	0.554	0.563	0.058	0.525	0.535	0.059
RRR	0.223	0.318	0.318	0.230	0.344	0.354

including a class with only three cases, is treated as a diagnostic counterexample: lower leakage accompanies lower clean accuracy. It does not carry the evidential weight of the primary CUB and Pet benchmarks.

Seven dataset-tailored simulations further probe specific sensitivities (Table IX). CUB uses twilight and camouflage backgrounds; Pet uses warm household lighting and a checkerboard background; ISIC adds hair or ruler artifacts outside the lesion and separately reduces lesion contrast while preserving surrounding skin. FCAL improves both CUB and Pet scenario means, whereas the ISIC hair and ruler means are lower than CE. These are separate sensitivity tests, not natural-shift evaluations.

TABLE VII

OBSERVED FIXED-ARM TRAINING COST. TIME COVERS EPOCHS 6–50, INCLUDING VALIDATION AND EXCLUDING THE SHARED WARM-UP. RATIOS USE THE MATCHED FCAL RUN WITHIN EACH ARCHITECTURE; ENTRIES ARE MEDIANS [MIN, MAX]. PEAK MEMORY IS ALLOCATED CUDA MEMORY. FASTER-ISNET HAS ONE SUPPORTED ARCHITECTURE.

Method	Arch.	Median min	Time ratio	Memory ratio
FCAL (proposed)	6	5.8	1.00	1.00
CAMAL	6	6.4	1.09 [1.03, 1.15]	1.00 [1.00, 1.01]
GAIN-ext	6	10.4	1.82 [1.48, 2.23]	1.79 [1.62, 1.93]
RRR	6	31.9	4.49 [2.22, 29.67]	2.47 [1.95, 3.24]
Faster-ISNet	1	51.2	9.44	6.16

TABLE VIII

COMPONENT ABLATION ON CUB RESNET-50, SEED 42, AT EPOCH 20 OF THE SAME 50-EPOCH PARENT SCHEDULE. THE FIXED-CELL ROW RETAINS SIGNED SUPPORT BUT USES A THREE-CELL DILATION AFTER MASK PROJECTION.

Variant	Clean	BG mean	BG worst	Leak ↓
Fixed-cell tolerance	69.80	61.76	35.07	0.564
Without signed support	70.47	62.40	34.69	0.490
FCAL (full)	69.83	62.84	43.32	0.493

TABLE IX

DATASET-TAILORED SENSITIVITY TESTS, REPORTED SEPARATELY FROM THE COMMON 13-VIEW MEANS. ENTRIES ARE SIX-ARCHITECTURE MEAN ACCURACIES (PERCENT), SEED 42.

Dataset	Scenario	CE	FCAL
CUB	Twilight background	76.36	77.12
CUB	Camouflage background	72.03	73.70
Pet	Warm household light	89.19	89.80
Pet	Checkerboard background	87.23	87.73
ISIC	Hair outside lesion	56.83	56.33
ISIC	Ruler outside lesion	63.17	62.17
ISIC	Low-contrast lesion	35.00	36.67

V. DISCUSSION

FCAL provides an explicit training recipe for improving background robustness while preserving classification accuracy. The replicated CUB result establishes this operating point against CE; the remaining comparisons characterize its behavior relative to alternative explanation-supervision objectives. The following analysis considers the mechanism, empirical trade-offs, and limits of the evidence.

A. Comparative interpretation

The experiments do not yield a universal ranking. Under the fixed recipes, the classification endpoints generally favor FCAL, whereas common-CAM localization and tuned CNN worst-background accuracy favor CAMAL. The fixed FCAL recipe remains the primary configuration because validation-only coefficient selection did not improve it consistently. Table X summarizes these trade-offs within the scope of the reported protocol.

B. Why the objective can help

Fractional containment measures the location of positive map mass instead of its raw scale. The boundary tolerance avoids assigning identical input-space meaning to a cell in a coarse CNN grid and a cell in a finer ViT grid. Signed support supplies a complementary optimization signal when the positive map alone cannot express foreground support. Together these choices encourage object-centered features without requiring the network to match an entire binary silhouette. The component ablation and replicated foreground-only gains support this interpretation.

This interpretation fits the broader explanation-guided learning literature [13]. ISNet and Faster ISNet directly optimize LRP relevance [6], [7], whereas FCAL uses the inexpensive final-head map. CAMAL more directly rewards regional activation alignment [15], consistent with its stronger common-CAM localization in our study. These differences explain why selection should follow the application endpoint. FCAL is favored by the demonstrated classification results, whereas CAMAL is favored by localization and by the stronger CAMAL BG-worst pairings.

C. Why containment is not background independence

The stronger CUB CE ViT background baseline leaves less headroom than the CNN baseline. Token mixing, receptive fields, model size, pretraining, grid resolution, and the family-specific containment cap also differ. The comparison measures their combined effect under the declared recipes; it does not isolate convolution versus attention as the cause.

TABLE X
INTERPRETATION OF THE EMPIRICAL TRADE-OFFS. ENTRIES DISTINGUISH OBSERVED OUTCOMES FROM THE EVIDENCE SUPPORTING THEM.

Evaluation focus	Observed pattern	Evidence scope
FCAL coefficient selection	The declared fixed recipe is retained because validation-only selection of β does not improve the held-out endpoints consistently.	Six architectures; one seed per cell.
Classification under background change	Fixed FCAL has higher six-architecture means than fixed CAMAL on all four classification endpoints. The advantage is most consistent for clean and FG-only accuracy.	Seed 42; architecture-level point estimates.
Spatial alignment and tuned worst-case accuracy	CAMAL has higher common-map IoU and insertion AUC in both selection arms, and higher tuned CNN BG-worst accuracy on all three backbones.	Seed 42; common 500-image localization subset and validation-selected coefficients.
Generalization	CNNs show the clearest FCAL gains over CE; the average-pooled ViT results are mixed, Pet transfers positively, and the small ISIC check is negative.	Three CUB ResNet-50 seeds; otherwise one seed per cell.

Equation (1) is a logit decomposition, not a causal decomposition of input pixels. Convolutional receptive fields and transformer attention allow a feature at a foreground location to encode background information. The loss also targets positive evidence for the true class; negative background evidence and evidence for competing classes are not directly controlled.

There is a further algebraic limitation. Add the same vector $v \in \mathbb{R}^D$ to every classifier row: $w'_k = w_k + v$. Every logit gains the same image-dependent offset $v^\top (hw)^{-1} \sum_{ij} A_{ij}$, so softmax probabilities and predictions are unchanged. Nevertheless, the target map gains $v^\top A_{ij}$ and its containment or support can change. Thus the raw class map is not uniquely determined by the discriminative decision, even when its mean exactly recovers the logit contribution. Fractional normalization removes a scale dependence away from the numerical floor, not this common-class ambiguity. Our predictive conclusions therefore rest on held-out intervention accuracies, not on reading leakage as causal reliance.

The exploratory extensions in Appendix C improve some localization outcomes but fail to supply a consistently stronger predictive recipe, supporting retention of the original FCAL configuration.

D. Evidence strength and generalization boundary

Replicated finding: each of the three matched CUB ResNet-50 FCAL runs exceeds its CE counterpart on clean, BG-mean, BG-worst, and FG-only accuracy. The sample SDs describe training variation on one split, with limited precision from three repetitions. **Comparative evidence:** the architecture matrix and FCAL–CAMAL rankings use one seed per cell and are descriptive; they do not establish statistically confirmed method superiority. The component ablation is single-seed and short-horizon. Baseline comparisons cover CUB, with Faster-ISNet restricted to ResNet-50. Pet supplies external classification evidence; ISIC supplies only a small negative diagnostic. CAMAL is currently documented as an arXiv preprint. Its publication status does not alter the measured comparison, but the result pertains to the cited preprint formulation; RRR, GAIN-ext, ISNet, and Faster ISNet provide the archival baseline context.

Repeated exploratory development on the same validation split creates selection risk beyond within-run checkpoint sealing. No independent natural context-shift benchmark, partial-mask study, or mask-corruption experiment is included. Foreground annotations cost effort, may omit legitimate context, and need not coincide with all class-relevant evidence. Our exact map interface excludes general CLS-token and nonlinear pooled heads. Unconditional leakage can favor empty positive maps, and live-map and prediction-class decompositions remain unreported. The faithfulness comparison uses a common CAM rather than every method's native explainer.

The highest-priority confirmation is an independent multi-seed FCAL–CAMAL comparison with frozen recipes, followed by natural background shifts and incomplete/noisy-mask supervision. Table X therefore summarizes the observed trade-offs rather than establishing a universal method ranking.

VI. REPRODUCIBILITY

The study retains configurations, split manifests, checkpoint-selection records, deterministic variants, and per-model results. FCAL corresponds to implementation identifier `fcal_v2`. The source package includes the table generator, preserved inputs with SHA-256 hashes, bibliography, figures, paired seed differences, and corrected Pet tests. The tuning manifest binds all 25 winners to validation-selected coefficients; common-map evaluations record checkpoint and image-subset hashes. Full datasets and trained checkpoints are excluded from the manuscript package.

VII. CONCLUSION

FCAL couples fractional containment, signed foreground support, and resolution-aware tolerance with one backbone forward and unchanged inference. Its central result is a classification win: in three matched CUB ResNet-50 seeds, fixed FCAL improves clean, BG-mean, BG-worst, and FG-only accuracy over CE. In the single-seed six-architecture matrix, fixed FCAL also exceeds fixed CAMAL on the mean of every classification endpoint and leads on 6/6, 5/6, 4/6, and 6/6 architectures, respectively, at similar training cost. Under the declared protocol and at seed 42 for the six-architecture matrix, fixed FCAL therefore provides the stronger

classification result than fixed CAMAL. This advantage is endpoint-specific: CAMAL leads common-map localization and the reported tuned CNN worst-background endpoint.

Taken together, the results support the declared fixed configuration as the primary FCAL recipe; coefficient tuning remains a separate validation decision. The CNN/ViT contrast and small ISIC check define transfer boundaries around FCAL's demonstrated classification–robustness operating point.

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APPENDIX A
SEARCH GRIDS AND ACCOUNTING

For reproducibility, the one-dimensional grids are: FCAL support weight $\beta \in \{0.0125, 0.025, 0.05, 0.1, 0.2\}$; RRR gradient weight $\{10^{-5}, 3.16228 \times 10^{-5}, 10^{-4}, 3.16228 \times 10^{-4}, 10^{-3}\}$; GAIN-ext supervision weight $\{1, 3.16228, 10, 31.6228, 100\}$; CAMAL coefficient $\{0.1, 0.316228, 1, 3.16228, 10\}$; and Faster-ISNet heatmap mixing weight $\{0.5, 0.7, 0.8, 0.9, 0.95\}$. The declared fixed settings are respectively 0.05, 10^{-4}, 10, 1, and 0.7. Selection uses clean validation accuracy, with lower trial index breaking ties. Tables XI and XII report the selected settings and realized search charges. The same number of candidates and fidelity levels does not equalize hardware time or the expressiveness of the search spaces.

TABLE XI
VALIDATION-SELECTED COEFFICIENTS. THE COLUMNS ARE RESPECTIVELY THE FCAL SIGNED-SUPPORT WEIGHT, RRR GRADIENT WEIGHT, GAIN-EXT SUPERVISION WEIGHT, CAMAL ALIGNMENT COEFFICIENT, AND FASTER-ISNET HEATMAP MIXING WEIGHT. A DASH INDICATES AN UNSUPPORTED ARCHITECTURE; FASTER-ISNET SUPPORTS RESNET-50 ONLY IN THIS STUDY.

Architecture	FCAL	RRR	GAIN-ext	CAMAL	Faster-ISNet
ResNet-50	0.0125	10^{-5}	3.16228	3.16228	0.5
DenseNet-121	0.025	3.16228×10^{-5}	31.6228	0.316228	–
EfficientNet-B0	0.025	10^{-5}	3.16228	0.1	–
ViT-Tiny/16 avg	0.0125	10^{-5}	3.16228	0.1	–
ViT-Small/16 avg	0.025	10^{-5}	100	0.316228	–
ViT-Base/16 avg	0.05	3.16228×10^{-5}	100	1	–

TABLE XII
REALIZED VALIDATION-ONLY TUNING COST UNDER ALLOCATED-WALL-CLOCK GPU ACCOUNTING. ALL ATTEMPTS, INCLUDING FAILURES, ARE CHARGED. †FASTER-ISNET COVERS RESNET-50 ONLY.

Method	Scope	Attempts	Failures	GPU h	Failed GPU h
FCAL	six architectures	48	0	2.287	0.000
CAMAL	six architectures	48	0	2.326	0.000
GAIN-ext	six architectures	48	0	3.265	0.000
RRR	six architectures	48	0	11.566	0.000
Faster-ISNet†	ResNet-50 only	8	0	2.159	0.000
Total	declared support matrix	200	0	21.604	0.000

APPENDIX B
COMPLETE DATASET AVERAGES

Table XIII retains the full six-architecture diagnostic summary. These means complement the family and architecture breakdowns in the main text. Higher FG-corruption accuracy reflects resistance to foreground damage, which is distinct from retaining the foreground while changing its background. Background-only accuracy describes residual class signal under the specified region replacement, rather than furnishing a universal ranking criterion.

TABLE XIII
FIXED-RECIPE TEST ACCURACY AVERAGED OVER SIX ARCHITECTURES AT SEED 42. CUB AND PET ARE THE PRIMARY BENCHMARKS; ISIC IS A 100-IMAGE NEGATIVE CHECK. ENTRIES ARE PERCENTAGES EXCEPT LEAKAGE. BG/FG MEANS AVERAGE 13 INTERVENTIONS; WORST IS COMPUTED WITHIN EACH ARCHITECTURE BEFORE AVERAGING. ARCHITECTURE MEANS ARE DESCRIPTIVE.

Dataset	Method	Clean	BG mean	BG worst	FG mean	FG worst	FG only	BG only	Leak ↓
CUB	CE	79.48	72.54	50.40	37.42	2.49	72.39	5.25	0.685
CUB	FCAL (proposed)	80.05	73.75	55.05	37.06	2.13	73.31	4.50	0.473
Pet	CE	89.52	86.81	74.99	46.85	3.26	86.59	4.03	0.292
Pet	FCAL (proposed)	90.03	87.48	76.29	47.46	2.97	88.94	3.54	0.191
ISIC	CE	66.50	49.65	34.83	37.00	15.83	42.50	23.33	0.441
ISIC	FCAL (proposed)	64.83	49.82	33.83	35.76	15.33	46.17	21.67	0.257

APPENDIX C
EXPLORATORY OBJECTIVE EXTENSIONS

This appendix reports completed exploratory follow-ups to explain why the retained formulation is not replaced by a later development version. The internal identifiers below identify archived candidates, not additional headline methods. These studies reuse CUB during development and do not provide independent confirmation of the main claims.

A. Regional classification and buffered suppression

The foreground-pooling candidate (V2.7) adds classification from features pooled inside the foreground, using the existing linear head and a fixed weight of 0.02. The buffered candidate (V2.7′) instead erodes the foreground and excludes a dilated foreground from the background pool, using the input-space radius in Equation (3). Each pooled vector is classified by the shared head. Its target-versus-rest margin is the target logit minus the log-sum-exp of non-target logits. Softplus of the negative foreground margin promotes target classification; softplus of the background margin suppresses it. Their sum receives an adaptive detached CE-to-regional-loss weight with factor 0.15 and bounds [0.05,0.30], under the existing ramp. Empty pools are excluded from their corresponding mean. Both candidates preserve the full FCAL objective and reuse one ordinary backbone forward.

Table XIV uses the complete two-seed audit with a common stochastic-intervention seed, rather than mixing earlier screening evaluations with different random backgrounds. V2.7 changes BG mean by $+1.06$ points at training seed 42 but -0.29 at seed 7. Buffered pooling changes it by $+0.30$ and $+0.28$ points; both image-bootstrap 95% intervals include zero (respectively $[-0.93, 1.51]$ and $[-0.85, 1.40]$ points). No individual accuracy comparison survives Holm correction across the 60 tests: two training seeds, two candidates, and 15 endpoints (clean, FG only, and 13 background views). These conditional image-level intervals and tests do not establish seed-level uncertainty.

TABLE XIV
EXPLORATORY CUB RESNET-50 VALIDATION RESULTS ON 1,200 IMAGES. THE SAME STOCHASTIC-INTERVENTION SEED 271828 IS USED FOR BOTH TRAINING SEEDS AND ALL CANDIDATES. THESE ARE VALIDATION RESULTS, SEPARATE FROM THE MAIN TEST COMPARISONS.

Seed	Objective	Clean	BG mean	BG worst	FG only	Leak ↓
42	FCAL	80.08	72.12	49.50	71.67	0.266
42	FCAL + FG pooling	80.00	73.18	52.75	73.42	0.256
42	FCAL + buffered pooling	80.33	72.42	48.17	71.83	0.230
7	FCAL	79.83	72.45	56.33	70.25	0.272
7	FCAL + FG pooling	78.75	72.15	56.67	71.17	0.268
7	FCAL + buffered pooling	78.67	72.73	57.17	71.83	0.215

The frozen V2.7 cross-CNN screen also regresses EfficientNet-B0 validation BG random-RGB accuracy by 4.08 points and FG-only accuracy by 7.34 points, despite a 0.59-point clean gain. DenseNet was stopped at epoch 10 on projected compute cost and has no eligible final quality comparison. These outcomes support retaining V2.7 as exploratory evidence rather than claiming it as a robust successor. The later buffered audit did not proceed to a third seed or official test evaluation.

B. Rival-class foreground support

The rival-support candidate (V2+) adds a foreground margin between the target map and the two highest-logit non-target classes, selected with detached logits. Each map is RMS-normalized separately before subtraction. For each rival, the foreground average of the resulting difference after $\tanh$ is encouraged toward 0.25 using a softplus with scale 0.25. The rival-averaged penalty has weight 0.025 and the existing training ramp. The normalized map difference is not an exact class-logit margin and does not remove the common-class ambiguity discussed in Section V.

After passing the frozen three-seed validation gates, this candidate was evaluated on the CUB test set (Table XV). Leakage decreases in all three seeds, but clean accuracy also decreases in all three. The paired BG-mean change is $+0.04 \pm 0.57$ points and the BG-worst change is $+1.88 \pm 4.13$ points, with the latter reversing sign at seed 7. This is a localization–prediction trade-off, not a demonstrated overall improvement. The smaller observed BG-worst SD for the candidate does not establish reduced population variance.

TABLE XV
EXPLORATORY CUB RESNET-50 TEST COMPARISON OVER SEEDS 7, 42, AND 123. ENTRIES ARE MEAN ± SAMPLE SD. THE LAST ROW SUMMARIZES PAIRED SEED DIFFERENCES; ACCURACY DIFFERENCES ARE PERCENTAGE POINTS. THE FCAL CONTROLS ARE THE SAME RUNS REPORTED IN THE MAIN REPLICATION.

Objective	Clean	BG mean	BG worst	FG only	Leak ↓
FCAL	78.49±0.16	72.40±0.38	51.67±4.00	72.51±0.87	0.281±0.003
FCAL + rival support	78.16±0.25	72.45±0.19	53.55±0.16	72.11±0.34	0.262±0.010
Δ rival support–FCAL	-0.33±0.28	+0.04±0.57	+1.88±4.13	-0.39±0.70	-0.019±0.012

The unchanged candidate transferred to ViT-Small at seed 42 improves validation clean accuracy by 0.33 points and lowers leakage by 0.240, but lowers BG mean by 1.01 points, BG worst by 0.92, and FG-only accuracy by 1.25. That candidate did not pass its validation gate and was not evaluated on the official test set. This additional result reinforces the distinction between improved map containment and predictive robustness.